

the drawback of higher heat generation. µModule from Linear Technology facilitates sharing of the heat-sink between digital microcontroller circuits and the module.

There is also another trend that is seen. Instead of adding heat-sinks and fans, the industry is trying to use the printed circuit board (PCB) itself to dissipate the heat generated. There are two developments to counter this problem. One is to use a material on the PCB that removes heat, and/or use multi-layer boards, and the other is to adopt multi-purpose packaging technologies.

Packaging plays a vital role.

Requirement for reduced package size has resulted in tiny lead-less or ball grid array (BGA) packaging with extremely small pitches. This requires reduced power consumption, leading to reduced voltages. Increases in thermal stresses on chips create a requirement for thermal pads, for effective heat dissipation. Heat from the chip goes into the power pad, from where it is transferred to the surface of the PCB and then dissipated.

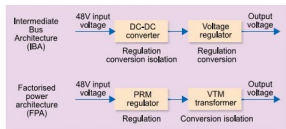


Fig. 2: Vicor's FPA: Factorising regulation, isolation and voltage transformation (Image courtesy: <http://powerblog.vicorpower.com/>)

MAJOR CONTRIBUTORS TO THIS REPORT



Balasubramanian,
Happest Minds
Technologies



Bhupinder Kathpalia,
senior application
engineer, Vicor Corp.



Esmond Wong,
VP business
development and
supplier marketing,
Arrow's global
components Asia-
Pacific business



Hemant Kamat,
chief technology
officer, Shalaka
Technologies



Kaustubh Karnataki,
chief engineer,
FluxGen Engineering
Technologies Pvt Ltd



Sanjay Dixit,
applications
engineer, Texas
Instruments India



Vikas Kumar Thawani,
analog applications
engineer, Texas
Instruments India

Quad Flat No Leads packaging is dominant. TO-XX packaging offers a wide range of small-pin-count packages for discrete parts like transistors or diodes. Smaller the package, higher is the professionalism required to solder the components.

This packaging trend not only satisfies the need to fit in maximum things into minimum space but also takes care of miniscule-size restrictions demanded by peripheral component interconnect express (PCIe) cards and wearable devices.

Making performance count

Optimising the performance of a chip within its limits of cooling, packaging and power supply capabilities is a mighty challenge. While dynamic voltage frequency scaling is one traditional method, MIC95410

7A load switch from Micrel permits power partitioning in systems, and is also capable of controlling power-up sequence using a ramp signal.

A stable and noise-free source. It is also necessary to make sure

that operation of a power-sourcing circuit is stable and noise-free. In a sensor based data-acquisition circuit, a slight change in output of the powering circuit can result in large variations in sensor calibration, resulting in erroneous data. Thus, precise voltage regulation of power-sourcing circuits is of prime importance for maintaining system integrity.

Improving efficiency. Vicor has come up with Factorised Power Architecture (FPA) to solve power-related performance issues. Enabling components are integrated power components called VI Chips. FPA aims at providing the highest degree of system flexibility, power density, conversion efficiency, transient responsiveness, noise performance and field reliability.

Design with caution. Design of the PCB also comes into prominence when inductive components like motors and solenoids are involved. These components tend to create spikes in current, and PCB parts have to be designed with optimum protection to handle this surge. There is also a need to build isolation into these solutions, separating the secondary from the effects of the primary winding of the transformer.

While trends in power-conversion components are not advancing as rapidly as, say, the Internet of Things or infotainment, tried-and-tested new developments are here to stay. Keep on the lookout for more. **EFY**

In the years to come, cars might drive design changes

In a white paper by Arthur Russell, principal applications engineer, Vicor Corp., the author says that in an electric vehicle scenario, most of the previously mechanically-powered systems are now implemented as electrical consumers. Power steering is now a high-power electrically-actuated drive-by-wire system. Air-conditioning uses a brush-less DC or 3-phase AC motor driven compressor and parking brakes use a motor to electrically actuate the callipers. These systems significantly increase the demand for safe-to-touch 12V power.